

EDUCATION

ENS Paris-Saclay

2022 - present

Master MVA

2024-2025

- Key courses : Optimal Transport, Theory of Deep Learning, Probabilistic Graphical Models

Master MPRI

2023-2024

- Key courses : Computer Vision, Parallel Computing , Advanced Deep Learning, Combinatorial Optimisation

L3 Mathematics and Computer Science

2022-2023

- Equivalent to Bachelor's degree in Mathematics and Computer Science.
- Final ranking : 2/29
- Key courses : Algebra, Statistical Learning, λ -calculus, Advanced Algorithmic

Lycée Aux Lazaristes

2020 - 2022

- Intensive two year course preparing for the highly competitive national entrance examination to the French Grandes Écoles.
- Key courses : Algebra, Calculus, Physics

Institution des Chartreux

2017 - 2020

Baccalauréat scientifique, mention Très Bien

- Equivalent to A levels in science subjects at grade A or above.

WORK EXPERIENCE

Cadenza Labs

Collaborator

2024-present

- Worked on Contrast-Pair Clustering for CCS-Style Methods. Paper published at EMNLP and at ICML Mech-Interp workshop [Laurito et al., 2024].
- Explored the emergence of linearly separable XORs in transformers latent space during training.

Internship

Emerging structures in Computational Graphs of Neural Networks

Goethe University, Frankfurt
March - September 2024

- Equivalent to a Master's Thesis. See the project here.
- Unsupervised behavioral control of models through graph algorithms.
- Introduced new type of nodes for mechanistic analyses of neural network through functional similarity between features.

Supervised Program for Alignment Research

2023-2024

- Discovered Cadenza Labs through SPAR.

Federated Learning with Untrusted Servers

INRIA Lille, 2023

- Equivalent to a Bachelor's Thesis.
- Learned cryptography applied to machine learning.
- Explored the paradigm of federated learning and it's various challenges. I particularly enjoyed the question of how to merge models trained independently in parallel.

ML4G - EffiScienceResearch

Villars Les Dombes, 2022

- ML bootcamp organised by a group dedicated to AI safety formations.
- Learned to use PyTorch through various introductory projects to interpretability and alignment.
- Most notably learned the basics of mechanistic interpretability, which is the subject I'm interested for future research.

Wolfram Research

Oxford, 2019

- Organized for pupils who stood out during the 2019 Mathematics Olympiads, in which I was awarded the 9th place in Lyon during high school.

Volunteering

Ailleurs et Autrement

Neyron, 2016-2019

- Organisation dedicated to organizing trips around the world for mentally or physically disabled people.

SKILLS

Programming

- Python / PyTorch (proficient)
- C/C++ (proficient)
- Ocaml (highly familiar)
- L^AT_EX (proficient)
- x86 – 64 *assembly* (basic knowledge)

Leisure and Interests

- Ice creams (expert)
- Petanque (proficient)
- Sleeping (beyond legend)

REFERENCES

Walter Laurito, Sharan Maiya, Grégoire Dhimoïla, Owen Ho Wan Yeung, and Kaarel Hänni. Cluster-norm for unsupervised probing of knowledge. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14083–14112, Miami, Florida, USA, November 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.emnlp-main.780>.